

11

Surface Areas and Volumes



The surface area and volume of a swimming pool are calculated in order to determine how much water has to be added. The area or region that an object's surface occupies is known as its surface area, whereas volume refers to an object's usable space.



What's Inside

Topic(s)

- ✓ Topic-wise Notes
- ✓ Related NCERT Qs
- ✓ Caution Points
- ✓ Case-based Examples

Questions

- ✓ New Pattern Qs
- ✓ All Objective + Subjective Qs
- ✓ Diksha + Exemplar Qs
- ✓ Detailed Explanation

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| TOPIC 1 |

INTRODUCTION TO SURFACE AREA AND VOLUME

Learning Objectives

- Students will be learning about plane figures and solid figures and differentiate them.

Learning Outcomes

- Students will be able to apply the formulae and find the surface areas and volumes of solid figures.

Real Life Application

When shifting to a new house, a tedious job comes into our pocket, i.e., painting, but nowadays wallpaper is more popular alternative as the process of pasting it, is relatively quick and inexpensive.



The first thing the worker does when they arrive at your home, is measure the room, take note of the wall's height and width, i.e., they determine the surface area of the walls before applying the wallpaper.

So, in order to wrap things, paint them, apply onto them, we need to determine their surface area. Let's look at how to calculate the surface area and volumes of various 3-D shapes.

We are surrounded by regular solids all around us, but until now we have dealt with figures that can be easily drawn on the pages of our notebook or on the

blackboard of our classroom. These are called plane figures. Plane figure is a geometry figure that has no thickness. We have learned about some of the plane figures like a rectangle, square and circle and their areas and perimeters. In this chapter, we shall study about some solid figures like right circular cone, sphere and hemisphere. These figures are three-dimensional figures which means that they have length, breadth and height. We shall also learn to find the surface area and volume of these figures.

The Surface Area of Solid Figures

The surface area of a solid figure, is the sum of the areas of all faces in that solid shape. It is always measured in square units. Hence, surface area is also known as total surface area.

Important

- The curved surface area of the solid figure is an area of only a curved surface, leaving the circular top and base or the outer area of the figure we can touch.
- The general formula of $CSA = \text{perimeter of base} \times \text{height of figure}$.
- Total surface area = $CSA + \text{Area of bases}$.

Caution

- The perimeter of the base is required to calculate the curved surface area.
- 3-D figures have curved surface area whereas 2-D figures have lateral surface area.

Volume of Solid Figures

Volume of a solid figure is simply defined as the amount of space occupied by any three-dimensional solid. It is always measured in cubic units.

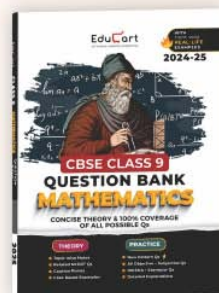
Important

- Volume of a figure generalised formula = $\text{Area of base} \times \text{height of the figure}$.



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OBJECTIVE Type Questions

[1 mark]

Multiple Choice Questions

1. If the radius of the sphere is $2r$, then its volume will be:

(a) $\frac{4}{3}\pi r^3$

(b) $\frac{8\pi r^3}{3}$

(c) $4\pi r^3$

(d) $\frac{32}{3}\pi r^3$

[NCERT Exemplar]

Ans. (d) $\frac{32}{3}\pi r^3$

Explanation: The radius of sphere = $2r$

[given]

The volume of sphere = $\frac{4}{3}\pi r^3$

Here, r is the radius of the sphere

New radius = $2r$

$$\therefore \text{Volume} = \frac{4}{3}\pi(2r)^3$$

$$= \frac{4}{3}\pi(8r^3)$$

$$= \frac{32}{3}\pi r^3$$

2. One large solid sphere of radius 9 cm is melted into three solid spheres having radius is the ratio 1 : 6 : 8. The value of the radius of a largest sphere is:

(a) 8 cm

(b) 16 cm

(c) 4 cm

(d) 12 cm

Ans. (a) 8 cm

Explanation: Let the radius are x , $6x$, $8x$. Then,

volume of the solid sphere = sum of the volumes of 3 solid sphere

$$\frac{4}{3}\pi R^3 = \frac{4}{3}\pi r_1^3 + \frac{4}{3}\pi r_2^3 + \frac{4}{3}\pi r_3^3$$

$$(9)^3 = (x)^3 + (6x)^3 + (8x)^3$$

$$9 \times 9 \times 9 = x^3 + 216x^3 + 512x^3$$

$$729 = 729x^3$$

$$x^3 = 1$$

$$x = 1$$

Radius of largest sphere = $8x = 8 \times 1 = 8$ cm

Fill in the Blanks

3. The volume of the cone is percent of volume of the cylinder having the same radius and height.

Ans. 33.33%

Explanation: Volume of cone = $\frac{1}{3}$ volume of cylinder i.e. $\frac{1}{3} \times 100 = 33.33\%$

True and False

4. The volume of a sphere is equal to two-thirds of a cylinder whose height and diameter are equal to the diameter of the sphere.

[NCERT Exemplar]

Ans. True

Explanation: Let the radius of the sphere and cylinder be r . Given, height of the cylinder = diameter of the base

$$h = 2r$$

radius of sphere = radius of cylinder = r

According to the given condition,

Volume of sphere = $\frac{2}{3} \times$ Volume of cylinder

$$\frac{4}{3} \pi r^3 = \frac{2}{3} \times \pi r^2 h$$

$$\frac{4}{3} \pi r^3 = \frac{2}{3} \times \pi r^2 \times 2r$$

$$\frac{4}{3} \pi r^3 = \frac{4}{3} \pi r^3$$

Hence, volume of the sphere is equal to two-thirds of the volume of a cylinder.

Assertion and Reason (A-R)

Direction for the given question: A statement of Assertion (A) is followed by a statement of Reason (R).

Choose the correct option as:

5. Assertion (A): If the volume of two hemispheres are in the ratio 8 : 27, then the ratio of their radii are 2 : 3.

Reason (R): Volume of hemisphere is $\frac{4}{3} \pi r^3$.

Ans. (c) Assertion (A) is true but reason (R) is false.

Explanation: Let, volume of two hemispheres be V_1 and V_2

Given, $V_1 : V_2 = 8 : 27$



$$\frac{2}{3}\pi r_1^3 : \frac{2}{3}\pi r_2^3 = 8 : 27$$

$$r_1^3 : r_2^3 = 8 : 27$$

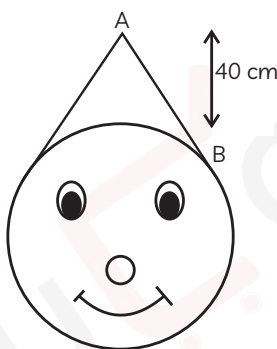
$$r_1 : r_2 = 2 : 3$$

CASE BASED Questions (CBQs)

[4 & 5 marks]

Read the following passages and answer the questions that follow:

6. On her birthday, Nisha brings a toy in the shape of a sphere and a cone attached to it on the other for her 5-year-old son. The spherical part's radius is 9 cm. The spherical and conical pieces have the same radii and the conical part is 40 cm tall.



(A)(A)The slant height of the conical part is:

- (a)41 cm (b)45 cm
(c)50 cm (d)42 cm

(B)The surface area of the speherical part, if the radius of sphere is 7 cm is:

- (a)600 cm² (b)308 cm²
(c)3988 cm² (d)616 cm²

(C)The surface area of the toy is:

- (a)2041 cm² (b)1775.71 cm²
(c)3988 cm² (d)980 cm²

(D)If Nisha wants to gift wrap the toy by using gift paper. Find the cost of paper used if 9 cm² paper ₹7.

- (a)₹1399.82 (b)₹1240.01
(c)₹1381.10 (d)₹1380

(E)The formula to find the surface area of toy is:

- (a) $\pi r^2 (r + l)$ (b) $\pi r (2r + l)$
(c) $2\pi r l$ (d) $\pi r l (l + r)$

Ans. (A) (a) 41 cm

Explanation: AB is the slant height of the cone.

Here, $r = 9$ cm, $h = 40$ cm

We know,

$$\begin{aligned}l^2 &= r^2 + h^2 \\l^2 &= (9)^2 + (40)^2 \\l^2 &= 81 + 1600 \\l^2 &= 1681 \\l &= \sqrt{1681} \\l &= 41 \text{ cm}\end{aligned}$$

Hence, the slant height of the cone is 41 cm.

(B)(d) 616 cm^2

Explanation: The curved surface area of sphere $= 4\pi r^2$

Here, $r = 7$ cm

Curved surface area $= 4\pi r^2$

$$\begin{aligned}&= 4 \times \frac{22}{7} \times 7 \times 7 \\&= 616 \text{ cm}^2\end{aligned}$$

(C)(b) 1775.71 cm^2

Explanation: Let r cm be the radius and h cm be the height of the conical part.

Then $r = 9$ cm, $h = 40$ cm

Also, $l = 41$ cm

[as part A]

Now,

The surface area of the toy = surface area of spherical part + curved surface area of the conical part

$$\begin{aligned}&= 616 + \pi rl \\ \Rightarrow &616 + \frac{22}{7} \times 9 \times 41 \\&= 616 + 1159.71 \\&= 1775.71 \text{ cm}^2\end{aligned}$$

(D)(c) ₹1381.10

Explanation: To wrap the toy we need to focus on the curved surface area of the toy.

As in the last question, we calculated the curved surface area of toy

The curved surface area of toy

$$= 1775.71 \text{ cm}^2$$

Cost of wrapping paper for $9 \text{ cm}^2 = ₹7$



Cost of wrapping paper for 1775.71 cm^2

$$= \frac{1775.71 \times 7}{9}$$

$$= ₹1381.10$$

The total cost of wrapping paper is ₹1381.10

(E)(b) $\pi r (2r + l)$

Explanation: The toy has a spherical part as well as conical part, so the formula used to find the surface area of toy is $\pi r (2r + l)$.

VERY SHORT ANSWER Type Questions (VSA)

[1 mark]

7. How many cubic centimeters of water can a conical vessel of base diameter 42 cm and slant height 29 cm hold? [Diksha]

Ans. Given, base diameter = 42 cm

radius, $r = 21 \text{ cm}$

Slant height,

$$l = 29 \text{ cm}$$

We know, $l^2 = r^2 + h^2$

$$h^2 = l^2 - r^2$$

$$h^2 = (29)^2 - (21)^2$$

$$h^2 = 841 - 441$$

$$h^2 = 400$$

$$h = \sqrt{400}$$

$$h = 20 \text{ cm}$$

$$\text{Volume of cone} = \frac{1}{3} \pi r^2 h$$

$$= \frac{1}{3} \times \frac{22}{7} \times 21 \times 21 \times 20$$

$$= 9240 \text{ cm}^3$$

SHORT ANSWER Type-I Questions (SA-I)

[2 marks]

8. A hemispherical bowl made of brass has an inner radius 5.25 cm. What is the total cost of tin-plating it on the inside at the rate of 4 per cm^2 ? [Diksha]

Ans. The radius of hemispherical bowl = 5.25 cm

The surface area of hemisphere = $2\pi r^2$

$$= 2 \times \frac{22}{7} \times (5.25)^2$$

$$= 173.25 \text{ cm}^2$$

Cost of tin-plating $1 \text{ cm}^2 = ₹4$

Cost of tin-plating 173.25 cm^2

$$= ₹4 \times 173.25$$

$$= ₹693$$

SHORT ANSWER Type-II Questions (SA-II)

[3 marks]

9. The diameter of a sphere is decreased by 25% by what percent does its curved surface area decrease?

Ans. Let, d be the diameter of the sphere.

$$\therefore \text{Surface area} = 4\pi r^2$$

$$\Rightarrow 4\pi \left(\frac{d}{2}\right)^2 = \pi d^2$$

On decreasing its diameter by 25% the new diameter

$$\begin{aligned} d_1 &= d - \frac{25d}{100} \\ &= \frac{75}{100} \times d \\ &= \frac{3d}{4} \end{aligned}$$

$\therefore$ The new surface area

$$\begin{aligned} &= 4\pi \left(\frac{d_1}{2}\right)^2 = 4\pi \left(\frac{9d^2}{64}\right) \\ &= \frac{9\pi d^2}{16} \end{aligned}$$

Decrease in surface area

$$\begin{aligned} &= \pi d^2 \left(1 - \frac{9}{16}\right) \\ &= \pi d^2 \left(\frac{7}{16}\right) \end{aligned}$$



∴ Percentage decrease in surface area

$$= \left(\pi d^2 \times \frac{7}{16} \times \frac{1}{\pi d^2} \times 100 \right) \%$$

$$= \left(\frac{700}{16} \right) \%$$

$$= 43.75\%$$

LONG ANSWER Type Questions (LA)

[4 & 5 marks]

10. The volume of a sphere is 4851 cm^3 . How much should its radius be reduced so that its volume become $\frac{4312}{3} \text{ cm}^3$?

[Delhi Gov. QB 2022]

Ans. We have,

$$\text{Volume of Sphere} = 4851 \text{ cm}^3$$

We know that, the volume of Sphere

$$= \frac{4}{3} \pi R^3$$

$$\text{So, } \frac{4}{3} \pi R^3 = 4851$$

$$\frac{4}{3} \times \frac{22}{7} \times R^3 = 4851$$

$$R^3 = \frac{3 \times 4851 \times 7}{4 \times 22}$$

$$R = \sqrt[3]{1157.625}$$

$$R = 10.5 \text{ cm.}$$

Now, for volume of the sphere = $\frac{4312}{3} \text{ cm}^3$

$$\frac{4}{3} \pi r^3 = \frac{4312}{3}$$

$$\pi r^3 = \frac{4312}{4}$$

$$r^3 = 1078 \times \frac{7}{22}$$

$$r = \sqrt[3]{343}$$

$$r = 7$$

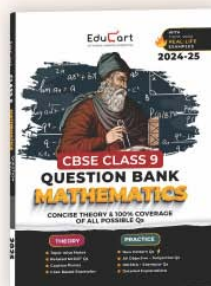
Now, change in radius

$$R - r = 10.5 - 7 = 3.5 \text{ cm}$$



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